Exposure-weighted versus lagged-treatment carryover specifications under differential-correlation biomarker interactions: a factorial simulation study

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Abstract

Background. Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Under the multivariate-normal (MVN) differential-correlation architecture (Architecture B), in which the biomarker-treatment interaction is encoded in a treatment-state-dependent correlation, a companion study found that carryover – residual drug effect that decays after discontinuation – costs up to roughly 31% of power in relative terms. The analyst must still decide how to represent residual exposure in the linear predictor, and it is not established which representation is most robust to the (typically unknown) pharmacokinetic decay shape.

Methods. We compare two analysis-model specifications for the drug-exposure regressor under Architecture B: an exposure-weighted continuous indicator that assumes a parametric decay and half-life (A2, the current default), and the binary on-drug indicator augmented by a lagged just-off-drug term that estimates carryover non-parametrically (A3, the classical crossover device). We embed them in an ADEMP-structured factorial simulation[1] crossing two data-generating process (DGP) decay forms (exponential, Weibull) with the two specifications, under three trial designs, two sample sizes, and three interaction strengths (216 cells, 500 replicates each), plus four single-axis sensitivity blocks and a cluster-robust recalibration. Every performance measure is reported with its Monte Carlo standard error (MCSE).

Results. A2 attains the highest power only in the two designs with a blinded-discontinuation phase (Hybrid and OL+BDC), where its advantage over A3 is about 10 percentage points (0.860 vs 0.770 at the reference cell). Under the classical crossover (CO) design A2 is markedly inferior (0.488 vs 0.830 for A3, which tracks the binary indicator closely). Where A2 leads, the lead is robust to decay-shape and half-life mis-specification and to autocorrelation strength, and a paired-McNemar high-precision rerun confirms it is real at the highest-leverage cell (+7.2 points, $p = 1.2 \times 10^{-7}$), though the two specifications are statistically tied at three of four other rerun cells. The model-based interaction test is mildly conservative (mean null rejection 0.039), an artefact of the working covariance that a CR2 cluster-robust standard error removes without disturbing the ranking.

Conclusions. Under Architecture B, the exposure-weighted predictor A2, reported with a cluster-robust standard error, is recommended for the Hybrid and OL+BDC designs; the lagged-treatment specification A3 is preferable, or at least not worse, under a crossover design. Across all conditions, anticipated dropout governs power far more than the choice between the two specifications, and should dominate design effort accordingly. This manuscript restricts the grid to Architecture B and to the exponential and Weibull decay forms, and does not evaluate the binary on-drug specification

(A1) or Architecture A (mean moderation); the full-scope evaluation, including those comparisons, is in the companion report.Rmd / report-slim.Rmd manuscripts.

1 Introduction

1.1 A short primer on aggregated N-of-1 trials

[1]

Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers at far smaller sample sizes than parallel-group trials.

- A single N-of-1 trial crosses one patient on and off treatment repeatedly; pooling many such series gives population-level inference.
- The estimand is the biomarker-treatment interaction: whether a baseline covariate B_i modifies the treatment effect.
- Three designs (crossover (CO); open-label plus blinded-discontinuation (OL+BDC); Hybrid) differ in how on- and off-drug occasions are arranged.

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In a single N-of-1 trial one patient is repeatedly crossed between active treatment and a comparator over a sequence of measurement occasions; the within-patient on-versus-off contrast estimates that patient's individual treatment effect. An *aggregated* N-of-1 trial pools many such single-patient series in a mixed model, trading some within-patient resolution for population-level inference and far smaller sample sizes than a parallel-group trial[2]. The estimand of interest is the biomarker-treatment interaction, i.e. whether a baseline covariate B_i modifies the treatment effect. We study three designs that differ in how on- and off-drug occasions are arranged: a classical crossover (CO), an open-label plus blinded-discontinuation design (OL+BDC), and a Hybrid N-of-1 design; these are defined in §3.

[2]

Carryover – residual drug effect that decays after discontinuation – contaminates off-drug occasions and attenuates the interaction test.

- The drug effect fades over days to weeks rather than switching off, so off-drug occasions carry residual exposure.
- Treating them as cleanly off-drug mis-specifies the design matrix.
- We ask which of two analyst-side exposure representations is most robust to the resulting attenuation.

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The inferential nuisance specific to this setting is **carryover**. When the drug is discontinued its pharmacological effect does not switch off but decays over days to weeks, so off-drug occasions carry residual exposure. Treating them as cleanly off-drug mis-specifies the design matrix and attenuates the interaction test. The question we take up is which of two analyst-side representations of residual exposure best recovers the lost power.

1.2 What is already known, and the gap

[3]

Under the differential-correlation architecture (B), carryover costs roughly 31% of power in relative terms, and the choice of analysis specification for representing residual exposure has not been evaluated against decay-form uncertainty.

- Architecture B encodes the biomarker interaction in a treatment-state-dependent correlation, which carryover erodes directly.
- A companion study fixed the decay shape at exponential and used a single analysis specification.
- This manuscript restricts to Architecture B, where the carryover problem is largest, and asks which specification is most robust when the decay shape is uncertain.

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A companion paper[3], extending Hendrickson[2], shows that under **Architecture B** (multivariate-normal differential correlation) the interaction lives in the treatment-state-dependent covariance $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, which carryover erodes directly: at a half-life of one week, power falls by up to roughly 31% in relative terms, strongly design-dependent (largest under OL+BDC, negligible under CO). Under the alternative Architecture A (direct mean moderation), the corresponding loss is only 1-3%; that architecture, and the third combined Architecture C, are outside the scope of this manuscript, which restricts to Architecture B, the architecture in which the carryover problem is largest and in which Hendrickson's original results were obtained. That companion work, however, fixed the decay shape at exponential and evaluated a single analysis-model specification throughout, leaving open which analyst-side representation of residual exposure is most robust when the true decay shape is uncertain.

1.3 This paper

[4]

We compare two analyst-side exposure representations – exposure-weighted (A2) and lagged-treatment (A3) – across decay forms under Architecture B.

- A2 commits to a parametric decay and half-life; A3 estimates carryover non-parametrically via a lagged term.
- Each is crossed against exponential and Weibull DGP decay forms.
- The question is whether one specification is recommendable or the choice is design-dependent.

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We evaluate two analyst-side representations of drug exposure in the linear predictor: an exposure-weighted continuous indicator that commits to a parametric decay form and half-life (**A2**, the current default), and the binary on-drug indicator augmented by a lagged just-off-drug term that estimates carryover non-parametrically (**A3**, the classical crossover device of Jones and Kenward[4]). We cross these against the exponential and Weibull data-generating decay forms and ask whether a single specification can be recommended or whether the choice is design-dependent. The study follows the ADEMP reporting framework of Morris, White, and Crowther[1], with every performance measure accompanied by its MCSE. Section 3 gives the simulation design in ADEMP order; Section 4 the results; Section 5 the practical implications for trialists.

2 Simulation design

2.1 Aims

[5]

The aim is to identify, under Architecture B, which of A2 and A3 is the more robust carryover-mitigation strategy across DGP decay forms and trial designs.

- Quantify the power cost of analysis-model carryover mis-specification under Architecture B.
- Identify which of A2 and A3 is more robust across the DGP decay-form grid and across trial designs.
- Verify whether the ranking survives half-life mis-specification and other single-axis perturbations (Tier 2).

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The aim of this study is to quantify, under Architecture B, the cost of analysis-model carryover mis-specification in aggregated N-of-1 and crossover trials, and to identify which of the two carryover-mitigation strategies, A2 or A3, is more robust across the DGP decay-form grid and across trial designs. A secondary aim, addressed in the Tier 2 sensitivity blocks, is to verify that the ranking survives perturbation of autocorrelation strength, dropout, half-life mis-specification, and interaction effect size.

2.2 Data-generating mechanism

[6]

This manuscript retains only Architecture B (MVN differential correlation); the biomarker and biological response are jointly MVN with correlation decaying off-drug.

- $\mathrm{Cor}(B_i, BR_{it}) = c_{bm}\,\phi(t_{sd})$, where ϕ is the decay form.
- Architecture A (mean moderation) and the combined Architecture C are out of scope here; see report.Rmd for the cross-architecture comparison.
- Two decay forms for ϕ are evaluated: exponential and Weibull. The linear form is dropped from this manuscript's scope.

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In Architecture B, the biomarker and biological response are jointly drawn from a multivariate normal with treatment-state-dependent correlation $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form and t_{sd} is time since drug discontinuation. Architecture A (direct mean moderation) and the combined Architecture C are outside the scope of this manuscript; the full grid comparison across architectures is reported in the companion report.Rmd / report-slim.Rmd manuscripts. In both architectures the decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps t_{sd} to a residual-effect multiplier. We evaluate two forms here, each parameterized to match a common half-life $t_{1/2}$; the linear form used in the full-scope manuscripts is dropped from this narrower evaluation.

2.2.1 Exponential decay

$$\phi_{\exp}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

[7]

The exponential form is the natural reference decay because it corresponds to first-order elimination kinetics.

- Used throughout the companion manuscript and implemented in `implementations/tidyverse/R/funct`
- Corresponds to classical pharmacokinetic first-order elimination.
- Retained as the reference against which the Weibull form is compared.

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This is the form used throughout the companion manuscript and currently implemented in `implementations/tidyverse/R/functions.R`. It corresponds to first-order elimination kinetics and is the default assumption in classical pharmacokinetics; we retain it here as the natural reference against which the Weibull form is compared.

2.2.2 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

[8]

The Weibull shape parameter k controls whether the elimination tail is heavier or lighter than exponential.

- $k < 1$: slower elimination with a prolonged low-concentration tail (evaluated at $k = 0.7$).
- $k > 1$: accelerated off-drug clearance via saturable clearance or active-transporter kinetics (evaluated at $k = 1.5$).
- $k = 1$ recovers the exponential form exactly.

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A shape parameter k governs whether the elimination tail is heavier ($k < 1$, slow elimination) or lighter ($k > 1$, accelerated off-drug clearance) than the exponential, which is recovered exactly at $k = 1$. We report results at $k = 0.7$, representing a drug with a prolonged low-concentration tail, and at $k = 1.5$, which captures drugs with saturable clearance or active-transporter kinetics. Together with $k = 1.0$ (numerically identical to the exponential form), this yields four decay-shape levels in the grid.

2.3 Estimand and analysis specifications

[9]

The estimand is the on-drug biomarker-response correlation c_{bm}; the null estimand $c_{bm} = 0$ is the Type I error row.

- Under Architecture B the interaction estimand is c_{bm}, dimensionless.
- Bias, empirical SE, and MSE are reported on the calibrated effect-size scale defined in `docs/19`.
- The null estimand for Type I verification is the $c_{bm} = 0$ row of the grid.

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Under Architecture B, the interaction estimand is the on-drug biomarker-response correlation c_{bm}, dimensionless. Bias and empirical standard error are reported on the calibrated effect-size scale (expected on-drug response difference per SD of biomarker at $t_{1/2} = 0$) defined in `docs/19-mean-moderation-implementation-notes.tex`; power and Type I error are reported on the native test-statistic scale. The null estimand for Type I error verification is $c_{bm} = 0$, realized by the $c_{bm} = 0$ row of the parameter grid.

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for both specifications. They differ only in the drug exposure predictor X.

2.3.1 A2: exposure-weighted (continuous predictor)

[10]

A2 uses a continuous exposure-weighted predictor that commits to a specific decay form and half-life, and is the current pmsimstats-ng default.

- $X = \mathrm{Dbc}_{it}$: decays off-drug according to the analyst's assumed form and half-life.
- Well-matched when the analyst's assumed form coincides with the true DGP form; mis-specified otherwise.
- Sensitivity of A2 to assumed-versus-true mismatch is evaluated in Tier 2 block S2.

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$X = \mathrm{Dbc}_{it}$, the continuous exposure-decayed predictor constructed to match the analyst's assumed decay form and half-life. This is the current pmsimstats-ng default specification. At on-drug timepoints $D_{bc,it} = 1$ regardless of the assumed half-life; at off-drug timepoints the predictor decays as $D_{bc,it} = \exp(-(\ln 2/\hat{t}_{1/2})\, t_{sd,it}) = (1/2)^{t_{sd,it}/\hat{t}_{1/2}}$, losing exactly half its value for every $\hat{t}_{1/2}$ weeks since discontinuation. When the analyst's assumed form coincides with the true DGP form the predictor is well-matched; when it does not, the predictor is mis-specified in a way that Tier 2 block S2 is designed to probe. The half-life assumption shapes the predictor values handed to `lme`; the model formula is unchanged for any $\hat{t}_{1/2}$, so A2 is effectively a different model for each assumed half-life even though the formula is identical.

2.3.2 A3: lagged-treatment (estimated carryover term)

[11]

A3 augments the binary indicator with a lagged-treatment term that estimates carryover magnitude without assuming a decay form, and is invariant to the analyst's assumed half-life by construction.

- $L_{it} = D_{i,t-1}(1 - D_{it})$ marks off-drug timepoints immediately following on-drug timepoints.

Table 1: Principal simulation grid (Architecture B only). Cell 1 is the matched reference: exponential DGP analyzed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification.

DGP decay	A2 exposure-weighted	A3 lagged-treatment
Exponential	cell 1 (matched)	cell 2
Weibull (k = 0.7)	cell 3	cell 4
Weibull (k = 1.0)	cell 5	cell 6
Weibull (k = 1.5)	cell 7	cell 8

- Carryover magnitude is estimated from the data; no functional form or half-life assumption is required.
- Follows the classical crossover-trial framework of Jones and Kenward [-@jones2014crossover].

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$X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ that marks off-drug timepoints following an on-drug timepoint. The carryover magnitude is estimated from the data rather than assumed a priori, so the specification does not require the analyst to posit a functional decay form. This approach follows the classical crossover-trial framework surveyed in[4]. One structural feature bears on interpretation: the lagged term L_{it} enters the model only as a main-effect nuisance covariate, and the biomarker interaction tested under A3 remains bm:D_{it}, the same coefficient as under the binary on-drug specification (A1, not evaluated in this manuscript). A3 is best understood as A1 with a carryover nuisance control rather than as a fundamentally different interaction model; unlike A2, it uses no half-life parameter and is therefore invariant to the analyst's assumed half-life by construction. This asymmetry – one specification indexed continuously by $\hat{t}_{1/2}$, the other not a member of that family at all – is the basis of the Tier 2 block S2 sensitivity contrast (§4.5).

2.4 Factorial grid and parameter settings

The principal comparison is a factorial of DGP decay form (four levels: exponential; Weibull at $k \in \{0.7, 1.0, 1.5\}$) against analysis-model specification (A2, A3), under Architecture B only:

Parameter	Values
Biomarker moderation (c_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.0, 1.5 (Weibull form only)
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	B (MVN differential correlation) only
Replicates per cell	500 (target); 50 for development

[12]

Restricting to Architecture B and dropping linear decay and A1 shrinks the grid to 216 cells, one-tenth of the full-scope manuscript's 540.

- Four decay-shape levels (exponential; Weibull at $k \in \{0.7, 1.0, 1.5\}$), architecture fixed at B.
- Grid $= 4 \times 3 \times 3 \times 2 \times 3 = 216$ cells (decay-shape $\times$ half-life $\times$ design $\times$ sample size $\times$ biomarker level).
- $c_{bm} = 0$ is the Type I error row; $t_{1/2} = 0$ is the no-carryover, best-case power reference.

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The 216-cell count follows the same nesting logic as the full-scope manuscript, with the architecture dimension fixed and the linear decay form dropped: the decay-shape and shape combinations number four (exponential; Weibull at $k \in \{0.7, 1.0, 1.5\}$), and the grid is $4 \times 3 \times 3 \times 2 \times 3 = 216$ cells (decay-shape $\times$ half-life $\times$ design $\times$ sample size $\times$ biomarker level, all at Architecture B). The $c_{bm} = 0$ condition serves as the Type I error check, targeting the nominal 5% level. The $t_{1/2} = 0$ condition removes carryover entirely and provides a best-case power reference at each $(N, \text{design}, c_{bm})$ combination, against which the power cost of non-zero carryover can be directly read off.

2.5 Performance measures and computing environment

For each cell we report Power, Type I error (at $c_{bm} = 0$), Bias and Empirical SE of the interaction estimate (on the calibrated scale), Coverage of the nominal 95% Wald CI, MSE, the non-convergence rate, and the MCSE of each measure, following Morris et al.[1, Tables 5-6].

[13]

$n_{sim} = 500$ keeps MCSE on power below 0.022; simulations reuse the tidyverse implementation and RNG strategy of the full-scope manuscript, restricted to Architecture B, non-linear decay, and A2/A3.

- Production: $n_{sim} = 500$, MCSE ≤ 0.022 at $p = 0.5$.
- Common random numbers across A2 and A3 within each cell enable paired contrasts.
- Data are the same production run summarised in `analysis/data/02-grid-summary.rds` (540 cells), filtered here to Architecture B, non-linear decay forms, and A2/A3.

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The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo standard error on power at $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$). This manuscript does not re-run any simulation; it filters the same production-scale grid summary used by the full-scope report.Rmd and report-slim.Rmd manuscripts (540 cells, `analysis/data/02-grid-summary.rds`, `implementations/tidyverse/` as the primary implementation, R 4.5.3, 128 minutes on eight cores, commit SHA `7018b59`, 2026-04-15) down to the 216-cell slice defined by Architecture B, non-linear decay forms, and the A2/A3 specifications. Within each cell, the same simulated dataset is used to fit both remaining specifications (common random numbers), so the A2—A3 contrast is estimated with reduced Monte Carlo variance relative to independent draws. Tier 2 sensitivity blocks and the S6 cluster-robust recalibration are filtered from the same shared outputs (`analysis/data/02-sensitivity-summary.rds`, `analysis/scripts/carryover-sensitivity/output/diag-s6-cr2.rds`); their designs are already anchored at Architecture B. Filtered figures are produced by `03-render-figures-extra-slim.R`, `04-sensitivity-figures-extra-slim.R`, and

Table 2: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). MCSEs are reported per Morris et al. (2019, Table 6).

t1/2	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non-conv
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

`07-type1-cr2-figure-extra-slim.R`.

3 Results

[14]

All reported results derive from the same 500-replicate production run as the full-scope manuscripts, filtered to Architecture B, exponential/Weibull decay, and A2/A3.

- MCSE on power ≤ 0.022 at $p = 0.5$; differences smaller than the relevant MCSE should not be over-interpreted.
- Coverage of nominal 95% Wald CIs is reported for the reference cell.
- Non-convergence rates are reported per cell in the tables that follow.

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3.1 Headline performance table

[15]

The headline table profiles the reference cell; bias, MSE, and coverage are calibrated to A2's target and are not directly comparable to A3's.

- Reference cell: Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$; non-convergence zero throughout.
- Bias/MSE/coverage are scored against A2's target $\theta_{\text{true}} = -c_{bm}\sigma_{BR} = -3.6$; A3 estimates a different coefficient (bm:D_{it}).
- A3's apparent bias and under-coverage reflect an estimand mismatch, not an estimator defect; compare only power and Type I error across specifications.

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3.2 Matched-decay baseline

[16]

The matched-decay baseline shows A2 power declining from 0.988 to 0.860 as carryover grows, reproducing Hendrickson's 0.82.

- Matched cell (exp DGP, A2, Architecture B, $N = 70$, $c_{bm} = 0.45$, Hybrid): $0.988 \rightarrow 0.948 \rightarrow 0.860$ across $t_{1/2} \in \{0, 0.5, 1.0\}$.
- All 500 replicates converged at every $t_{1/2}$.
- The $t_{1/2} = 1.0$ value matches Hendrickson's reported 0.82, confirming pipeline fidelity.

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Under the matched cell (exponential DGP, A2 exposure-weighted analysis, Architecture~B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate $\pm$ MCSE, $n_{sim} = 500$). All 500 replicates converged in the matched cell at every $t_{1/2}$. The value at $t_{1/2} = 1.0$ is consistent with Hendrickson's reported power of 0.82 for this cell[2], confirming that the simulation pipeline reproduces the baseline result.

3.3 Power by analysis specification

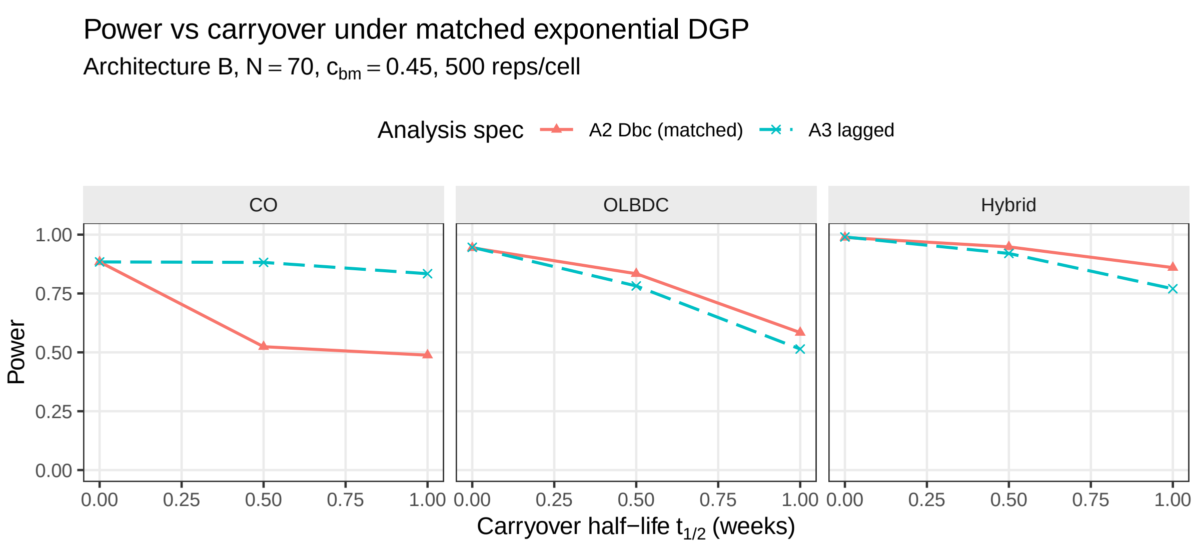

Figure 1: Power versus carryover half-life under the exponential DGP, Architecture B, stratified by analysis specification. A2 retains a modest advantage over A3 in the Hybrid and OL+BDC designs at $t_{1/2} = 1.0$ weeks, but is dominated by A3 under the crossover (CO) design.

[17]

The two specifications coincide at zero carryover and diverge as it grows, but A2's lead is confined to the two non-crossover designs.

- Both are near-identical at $t_{1/2} = 0$.
- A2's advantage emerges at $t_{1/2} = 0.5$ and grows in the Hybrid and OL+BDC designs.
- Under the CO design A2 is dominated by A3 (Section 5).

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The two analysis specifications yield near-identical power at $t_{1/2} = 0$, where there is no carryover to model, and diverge progressively as carryover half-life increases. A2 retains a modest but consistent advantage in the Hybrid and OL+BDC designs, where the advantage emerges at $t_{1/2} = 0.5$ weeks and grows through $t_{1/2} = 1.0$ weeks. This advantage does not appear under the crossover (CO) design, where A2 is in fact dominated (Section 5): in a crossover the within-subject correlation already carries most of the signal A2's decaying predictor is designed to recover, so A2 has nothing left to gain and its down-weighting of off-drug points only hurts.

3.4 Decay-form mis-specification

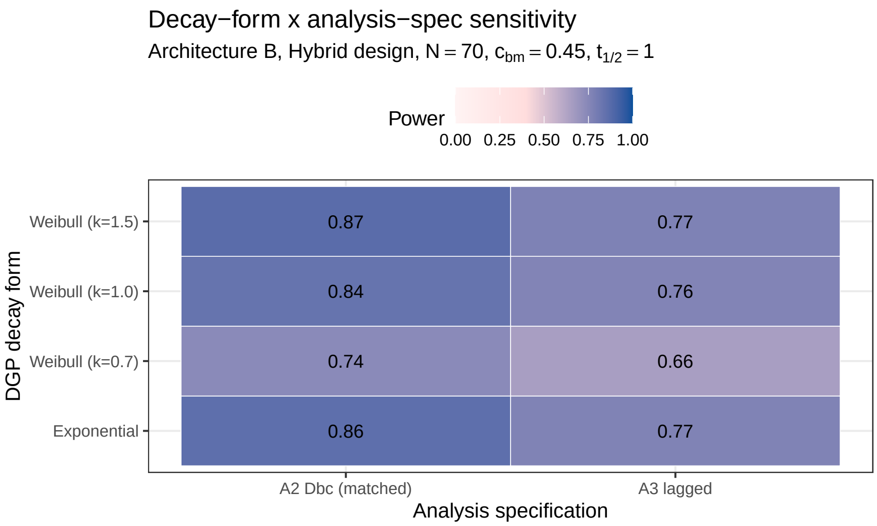

Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the heavy-tailed Weibull DGP ($k = 0.7$), A2's advantage over A3 narrows because its assumed exponential decay no longer matches truth.

[18]

The heatmap separates two mis-specification directions: wrong analysis given the DGP, and wrong assumed DGP.

- Matched reference (exp DGP × A2): power 0.860 ± 0.016.
- Same row, A3 column: penalty of the lagged specification under an exponential DGP.
- A2 column, other rows: cost of assuming exponential when the truth is Weibull.

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The heatmap cell at the intersection of the exponential DGP row and the A2 analysis column is the matched reference (power 0.860 ± 0.016). The same row's A3 column quantifies the power penalty of the lagged specification when the DGP is exponential; the A2 column's other rows quantify the cost of assuming exponential decay in the analysis when the true DGP is Weibull. Even under the heaviest-tailed Weibull DGP ($k = 0.7$), A2's advantage over A3 narrows but does not reverse; the Discussion (Section 5) takes up the practical interpretation.

3.5 Type I error control and cluster-robust recalibration

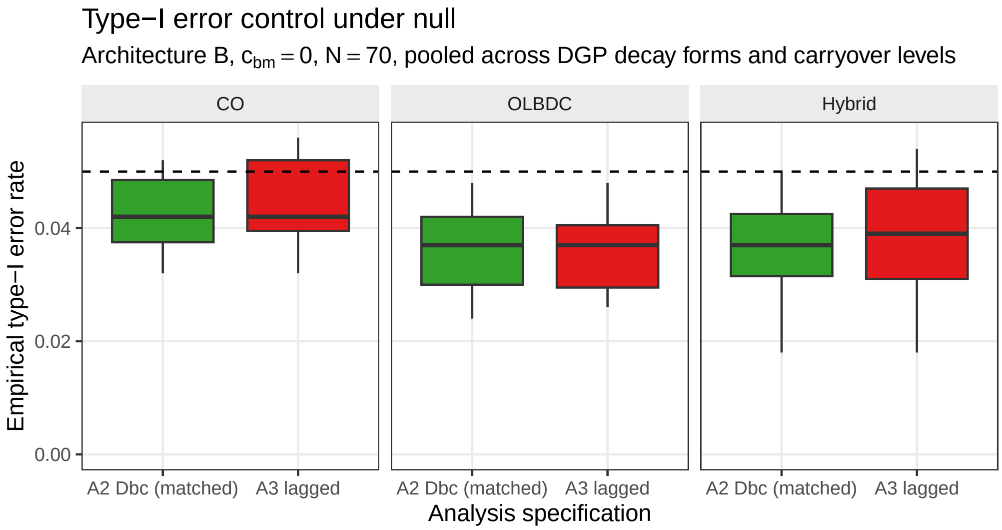

Figure 3: Empirical Type I error rates under the null ($c_{bm} = 0$), Architecture B, pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

[19]

Type I error is conservative, reflecting the corCAR1 working-covariance conservatism rather than inflation.

- Neither specification inflates the rate above nominal in any design.
- The companion calibration study traces the shortfall to a 6-10% over-estimated SE.
- Block S6's CR2 (below) restores nominal size without changing the ranking.

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Across the null cells, empirical Type I error is uniformly below the nominal 0.05 level for both specifications and all three designs. That shortfall is not sampling slack but the conservatism of the model-based interaction test that the companion calibration study[5] traces to the corCAR1 working covariance: its standard error is over-estimated by six to ten percent and the realized rejection rate falls correspondingly. The conservatism is uniform across specifications, which is why it leaves the comparison intact; the cluster-robust recalibration reported below (block S6) confirms that a CR2 cluster-robust standard error restores the rejection rate to nominal without disturbing the ranking. Table 3 makes the per-specification conservatism and its cluster-robust correction explicit at the null cell.

[20]

CR2 restores nominal size and recovers two to five power points without changing the A2-vs-A3 ranking, at the cost of far fewer degrees of freedom.

- Model-based null rejection sits near 0.03-0.04 across cells and specs; CR2 restores it to a mean near nominal.
- κ falls from 1.07-1.26 to about 1 under CR2.
- Validated only at Architecture B / Hybrid / $N = 70$; do not extend the reading to the CO design.

Table 3: Conservatism of the model-based interaction test and its cluster-robust correction, at the null cell ($c_{bm} = 0$, exponential DGP, $t_{1/2} = 1.0$ weeks, Hybrid design, $N = 70$, Architecture B, 3,000 replicates). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate; $\kappa > 1$ indicates a conservative test. Both specifications are conservative under the model-based test, most so the exposure-weighted A2 ($\kappa = 1.10$, Type I = 0.029); the CR2 bias-reduced cluster-robust standard error restores κ to near one and Type I to nominal (0.045 to 0.049) for both specifications.

Specification	κ model	Type I model	κ CR2	Type I CR2
A2	1.10	0.029	1.00	0.045
A3	1.05	0.039	0.98	0.049

Table 4: Cluster-robust recalibration of the exposure-weighted specification A2 at the reference cell and four stress cells (500 replicates, $c_{bm} = 0.45$, Architecture B). κ is the ratio of the mean standard error to the empirical standard deviation of the interaction estimate; a value above one is a conservative test. The A2−A1 columns (retained from the underlying diagnostic data; A1 itself is not otherwise reported in this manuscript) give the power gap between A2 and the binary reference under each inference, as a check that recalibration does not disturb the calibration mechanism. CR2 df is the mean Satterthwaite degrees of freedom for the cluster-robust test. Computed with the `clubSandwich` package.

Cell	κ model	κ CR2	Power model	Power CR2	A2−A1 model	A2−A1 CR2	CR2 df
Reference, $N = 70$	1.15	0.99	0.829	0.863	+0.080	+0.092	23
Small $N = 35$	1.13	0.96	0.558	0.590	+0.066	+0.086	12
High autocorr, $\rho = 0.9$	1.26	0.94	0.954	0.980	+0.052	+0.032	23
Half-life mis-spec	1.14	1.00	0.759	0.795	+0.010	+0.024	23
30% MCAR dropout	0.99	0.85	0.148	0.126	+0.010	+0.010	5

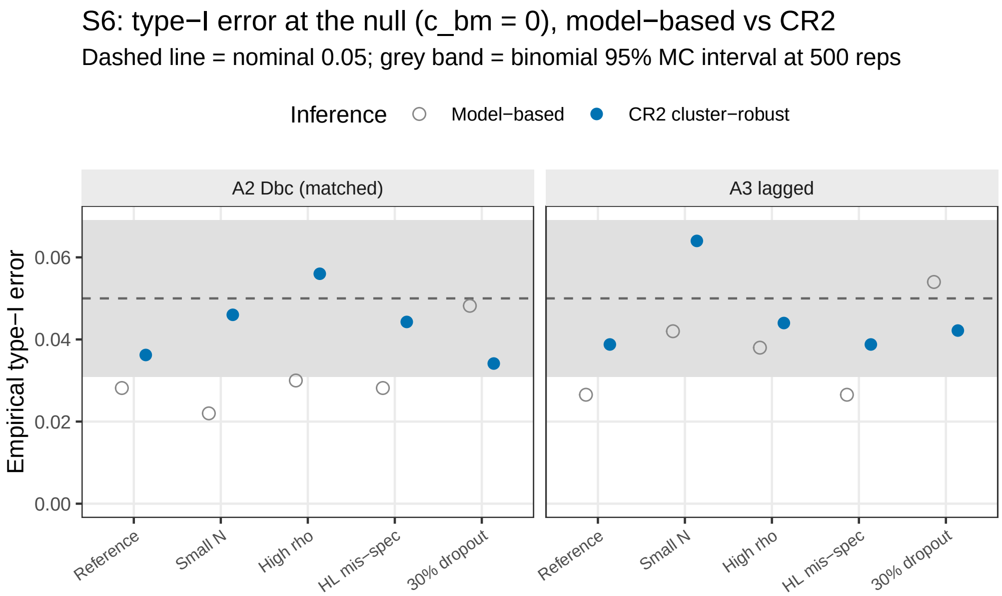

Figure 4: Empirical Type I error at the null ($c_{bm} = 0$) under model-based and CR2 cluster-robust inference, across the five S6 stress cells and the two specifications (Architecture B). The model-based test (open) is conservative (near 0.03); CR2 (filled) is restored to the nominal 0.05 (dashed), within the binomial Monte Carlo band (gray). 500 null replicates per cell.

rgt to complete.

Figure 4 shows the effect of recalibration on test size directly: the model-based null rejection rate sits below nominal across the five S6 cells and both specifications, while the CR2 test restores it to within the Monte Carlo band around 0.05. This is the precondition for the whole comparison – a test that is not correctly sized cannot be preferred on power alone. In the information-rich cells, the calibration ratio κ runs 1.07-1.26 under the model and is restored to about 1 under CR2, and the correctly-sized test recovers two to five points of power. The A2 advantage over the binary reference is held or slightly widened under CR2 (Table 4), so fixing the calibration does not change the ranking; this recalibration was validated only in the Architecture~B, Hybrid, $N = 70$ cells and should not be extended to the crossover design, where A2 does not lead.

3.6 Sensitivity analyses

[21]

Four Tier 2 blocks perturb one axis at a time around the reference cell.

- Reference: Architecture B, Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$.
- Each block varies one axis with the others held fixed.
- Blocks are reported separately, not pooled with Tier 1 or each other.

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3.6.1 S1: within-factor autocorrelation

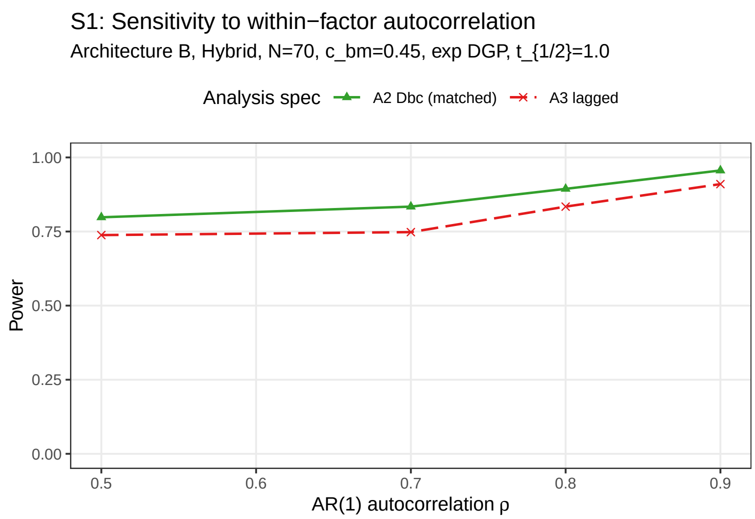

Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$, Architecture B.

[22]

S1: autocorrelation lifts both specifications equally, preserving A2's ≈ 0.06 advantage at every ρ.

- Power rises monotonically with ρ for both.
- A2's advantage holds near 0.06 across $\rho \in [0.5, 0.9]$.
- Autocorrelation and the specification act additively, not multiplicatively.

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Both specifications gain power monotonically with ρ. At $\rho = 0.5$, A2 power is 0.798 ± 0.018 versus A3 0.738 ± 0.020; at $\rho = 0.9$, A2 reaches 0.956 ± 0.009 versus A3 0.910 ± 0.013 (point estimate $\pm$ MCSE, $n_{sim} = 500$). The A2 advantage over A3, approximately 0.06 in absolute power throughout, is preserved at every AR(1) level: the interaction between autocorrelation strength and analysis specification is additive rather than multiplicative.

3.6.2 S2: analyst-truth half-life mismatch

[23]

S2: A2 keeps its advantage even when the assumed half-life is wrong by a factor of two or four.

- A2 averages 0.868 over eight assumed/true half-life cells (range 0.766-0.964).
- A3 is invariant to the assumed half-life (average 0.840).
- A2's gentle curvature makes it a defensible default under an approximately-known half-life.

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A2 is markedly robust to analyst-truth half-life mismatch. Across the eight S2 cells (DGP $t_{1/2} \in \{0.5, 1.0\}$ weeks crossed with analyst-assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), A2 averages 0.868, ranging from 0.766 to 0.964; per-cell MCSE ranges from 0.008 to 0.019.

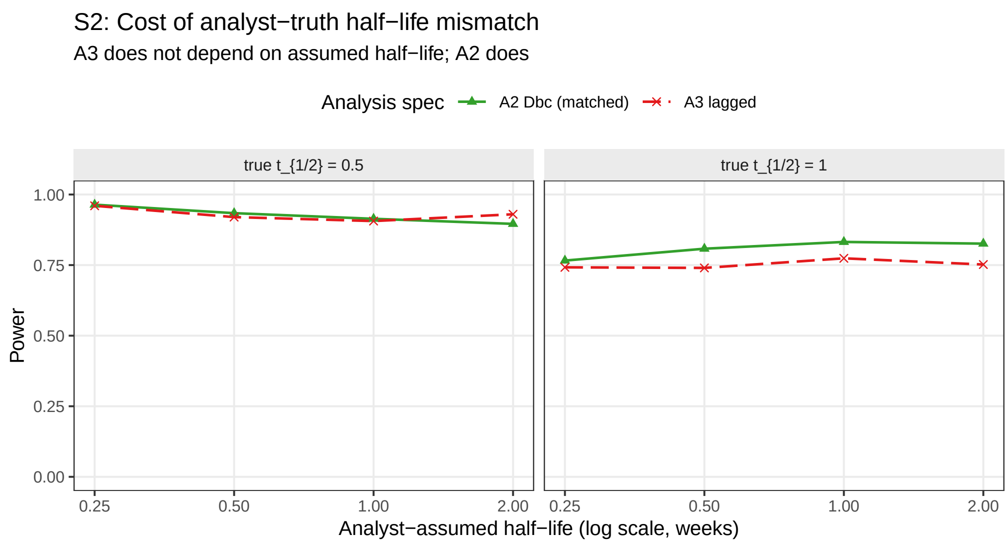

Figure 6: Power of A2 and A3 as the analyst's assumed half-life departs from the true DGP half-life. A3 is invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

A3 is invariant to the analyst-assumed half-life by construction, as it does not consume the parameter, and its power averages 0.840 across the same eight cells. The exposure-weighted predictor's curvature is sufficiently gentle that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage over A3 at either DGP half-life, suggesting A2 remains defensible even when the true pharmacokinetic half-life is only approximately known at the design stage.

3.6.3 S3: dropout

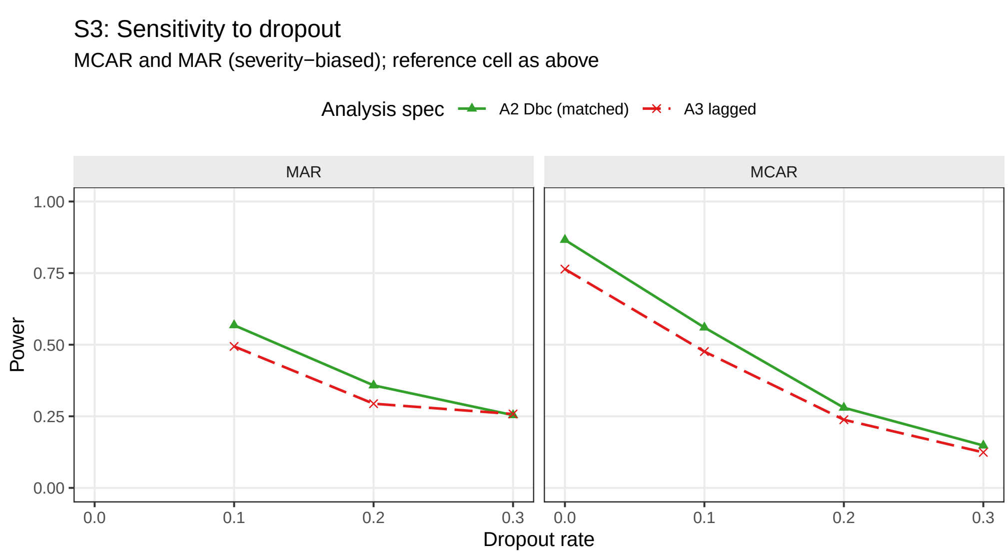

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms, Architecture B.

[24]

S3: dropout collapses power for both methods, and narrows A2's advantage as off-drug observations are dropped.

- At 30% MCAR both fall to about 0.12-0.15, effectively tied.
- The A2—A3 gap shrinks from ≈ 0.10 (no dropout) to ≈ 0.03 (30% MCAR).
- A2's signal lives in off-drug points, so dropping them erodes its edge; the advantage survives at dropout ≤ 0.20.

rgt to complete.

Both specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout $= 0$) gives A2 0.866 ± 0.015, A3 0.764 ± 0.019; at MCAR rate 0.30 these collapse to A2 0.148 ± 0.016, A3 0.124 ± 0.015. A2's advantage narrows under heavy dropout: the absolute A2—A3 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR, narrowing further under MAR at the highest rate (≈ 0.004 at rate 0.30). The pattern is consistent with A2's signal source residing in the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially dropped, A2's advantage contracts toward A3's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

3.6.4 S4: biomarker-moderation effect-size curve

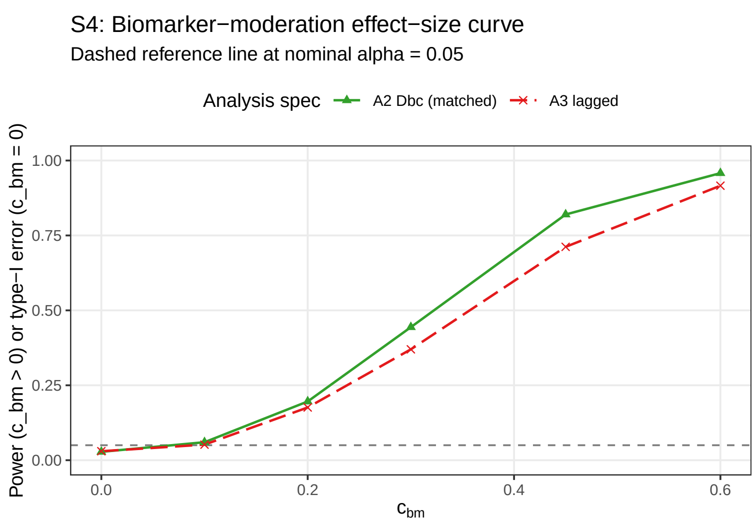

Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$ under Architecture B. The $c_{bm} = 0$ point is the null for Type I verification.

[25]

S4: the power curve is sigmoidal; A2's advantage peaks at intermediate effect sizes and vanishes toward the ceiling.

- Type I error at $c_{bm} = 0$ is conservative (0.028-0.030).
- A2—A3 peaks at +0.102 at $c_{bm} = 0.45$, shrinking to +0.042 at $c_{bm} = 0.60$.
- The advantage matters most for moderate biomarker-treatment interactions.

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The expected sigmoidal power curve is observed for both specifications. At $c_{bm} = 0$, empirical Type I error is below nominal at 0.028 ± 0.007 (A2) and 0.030 ± 0.008 (A3); power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A2 reaches 0.958 ± 0.009 and A3 0.916 ± 0.012. The A2 advantage is largest at intermediate effect sizes, $+0.082$ absolute at $c_{bm} = 0.30$ and $+0.102$ at $c_{bm} = 0.45$, shrinking to $+0.042$ at $c_{bm} = 0.60$ where both specifications approach ceiling. The advantage is therefore most practically consequential in designs targeting moderate biomarker-treatment interactions.

3.7 A higher-precision check

[26]

A high-precision paired-McNemar rerun finds A2 strictly ahead of A3 only at the highest-leverage cell ($+7.2$ points, $p = 1.2 \times 10^{-7}$); elsewhere the two are operationally tied.

- 24-cell OL+BDC rerun at 600 trials/cell, all converging.
- Paired McNemar (same trials): A2 beats A3 by 7.2 points at $N = 70$, $t_{1/2} = 1.0$; other three cells show $|\Delta| \leq 1.5$ points, $p \geq 0.18$.
- Type I error across the null cells stayed in $[0.003, 0.053]$.

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Because some Tier 1 gaps are small, a focused 24-cell grid (the OL+BDC design, $N \in \{35, 70\}$, true half-life $\in \{0.5, 1.0\}$) was rerun at 600 trials per cell, all converging; because A2 and A3 see the *same* trials, they can be compared with a paired **McNemar test**, more sensitive than treating the specifications as independent. Type I error across the twelve null cells fell in $[0.003, 0.053]$, so both specifications hold nominal alpha at the explored N and DGP half-life combinations. Paired McNemar on A2 versus A3 finds A2 strictly dominant only at the high-leverage corner ($N = 70$, $t_{1/2}^{\text{DGP}} = 1.0$ weeks, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the other three cells the paired difference lies between -0.5 and $+1.5$ percentage points with McNemar $p \geq 0.18$, and the two specifications are operationally tied at this resolution. Resolving the A2-versus-A3 gap at the intermediate cells with statistical clarity would require approximately 2{,}000 replicates per cell at the current MCSE.

4 Discussion

4.1 The main result, and where it breaks down

[27]

A2's advantage over A3 is real but conditional: confined to the Hybrid and OL+BDC designs, and reversed under the crossover design.

- Best cell (Hybrid): A2 beats A3 by ≈ 10 points (≈ 5 MCSEs).
- Under CO, A2 is much worse (0.488 vs 0.830 for A3).
- Mechanism: the crossover's within-subject correlation already carries the signal A2 targets, so A2 only loses there.

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The headline is that A2's advantage over A3 is real but **conditional**. At its best cell (Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$) A2 beats A3 by about 10 percentage points of power (0.860 vs 0.770) – about five MCSEs, so not luck. In the crossover design it is much *worse* (0.488 vs 0.830 for A3). There is a clean reason for the crossover failure: in a crossover the within-subject correlation already carries most of the signal that A2's decaying predictor is designed to recover, so A2 has nothing left to gain and its down-weighting of off-drug points only hurts. The ranking "A2 best" therefore holds only in the two non-crossover designs studied here.

4.2 What the robustness checks add

[28]

Where A2 leads, the lead survives all three robustness checks.

- Decay shape: A2 never drops below A3 even when the assumed form is wrong, up to the heaviest-tailed Weibull.
- Assumed half-life: a two- to four-fold error moves A2 only from 0.83 to 0.77, never below A3.
- Autocorrelation: lifts both equally, so A2's 5-6 point lead is ρ-independent.

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Three checks probe whether A2's win, where it exists, is fragile. **Decay shape (Fig. 2).** Even when the analyst wrongly assumes exponential decay but the truth is Weibull, A2 still does not drop below A3, within the decay-shape range tested here. **Assumed half-life (Fig. 6).** A3 does not use a half-life, so it is unaffected; A2 does, yet getting the half-life wrong by a factor of two or four moves its power only from about 0.83 to 0.77, never below A3. **Autocorrelation (Fig. 5).** Stronger serial correlation raises both methods by about the same amount, so A2's lead stays near 5-6 points regardless of correlation strength.

4.3 Dropout matters more than the method

[29]

Dropout collapses power for both methods equally, so it dominates the specification choice.

- At 30% dropout both fall to 0.12-0.25 and become indistinguishable.
- MAR (severity-related) dropout is slightly worse than MCAR, because the highest-signal patients leave first.
- Keeping patients in the trial buys more power than the choice between A2 and A3.

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The most sobering result is about *missing data*, not carryover. When patients drop out (Fig. 7), power collapses for both methods together: at 30% dropout both fall to 0.12-0.25 and become indistinguishable. A2 cannot protect against the loss; it only organizes whatever data remain. Dropout related to severity (MAR) is slightly worse than purely random dropout (MCAR), because the sickest patients – who carry the most signal – leave first. The practical message is blunt: keeping patients in the trial buys far more power than the choice between A2 and A3.

4.4 Scope, related work, and limits

[30]

This manuscript is a narrowed slice of the full-scope grid; its ranking should not be extrapolated to Architecture A, the binary specification A1, or linear decay without consulting the full-scope manuscript.

- Full grid (report.Rmd / report-slim.Rmd) shows A2's advantage does not extend to Architecture A, where A2 is marginally dominated throughout.
- A2 power 0.860 reproduces Hendrickson's 0.82; the Jones-Kenward expectation that shape-free A3 beats A2 under mis-specification is not supported here.
- Two further limitations: a single parameter set, and one-factor-at-a-time sensitivity checks.

rgt to complete.

This manuscript deliberately narrows the full-scope grid to Architecture B, non-linear decay, and the A2/A3 contrast, to bring a 30-page evaluation down to a size a reviewer can read at one sitting. The full-scope companion (report.Rmd / report-slim.Rmd) shows that the A2 advantage documented here does not extend to Architecture A (direct mean moderation), where A2 is marginally dominated in every design, nor is the binary on-drug specification A1 dominated by A2 outside the Hybrid/OL+BDC-under-Architecture-B cells studied here; readers who need the cross-architecture or A1 comparison should consult that manuscript rather than extrapolate from this one. Within the narrower scope, our matched-decay (A2) power at the headline cell, 0.860 ± 0.016, reproduces Hendrickson's[2] reported 0.82 for the same cell. The crossover tradition of Jones and Kenward[4] predicted that a shape-free lagged term (A3) should beat a parametric predictor (A2) when the decay shape is guessed wrong; our S2 result does not support this in the range we tested, because A2's penalty for a wrong half-life is too small for A3 to overtake it. Two further limitations bound these conclusions. First, the numbers come from one parameter set (calibrated to a prazosin/PTSD trajectory), so the *absolute* power values are illustrative; the *ranking* is the portable part, and even it is conditional on the design as shown above. Second, the sensitivity checks vary one factor at a time, so they can miss interactions between factors (for example, high autocorrelation combined with heavy dropout); the one joint check reported in the full-scope manuscript (block S5) showed no surprises.

5 Conclusions

1. **A2 helps, but only conditionally.** The exposure-weighted predictor gives the highest power – about 10 points over A3 – *only* in the Hybrid and OL+BDC designs, and the lead survives wrong decay-shape and wrong half-life assumptions there. Under the crossover design A2 has no advantage and A3 is at least as good. Where A2 is used, report it with a CR2 cluster-robust standard error (Block S6).

2. **A guessed half-life is fine (where A2 is used).** Getting the half-life wrong by a factor of two to four barely moves A2's power, so a rough pharmacokinetic estimate suffices.

3. **Dropout beats everything.** Power falls sharply as dropout rises (about half lost at 20% dropout), the same for both methods. Preventing dropout matters more than the choice between A2 and A3.

4. **The test is slightly conservative, and fixable.** Both methods reject a little less than the nominal 5% under the null; the CR2 cluster-robust standard error restores the correct rate while recovering a few points of power, without changing the ranking (Block S6).
5. **This is a narrowed slice.** Architecture A, the binary specification A1, and linear decay are outside this manuscript's scope; the full-scope grid in report.Rmd / report-slim.Rmd is the authority for those comparisons and should be consulted before generalizing beyond Architecture B.

Notes and reproducibility

The authors declare no conflicts of interest and received no specific funding. This manuscript is a scope-narrowed derivative of report-slim.Rmd: it re-filters, but does not re-run, the same production simulation. Cell-level summaries are at `analysis/data/02-grid-summary.rds` and `02-sensitivity-summary.rds`; filtered figures are produced by `03-render-figures-extra-slim.R`, `04-sensitivity-figures-extra-slim.R`, and `07-type1-cr2-figure-extra-slim.R` under `analysis/scripts/carryover-sensitivity/`. No simulation runs during knitting. The production simulation ran under R 4.5.3 and the document is rendered under R 4.6.0, both pinned via `renv.lock`.

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